

End-to-End Process • Plan-to-Produce (P2Pr)

Process Diagram

PCF 4.3 — BOMs + Routings · Pattern types color-coded · M3 HITL design intent at each step · Exception flows — colored left border

Pattern: **Decision (80% H)** · Knowledge (65% H) · Document (40% H) · Transaction (35% H) · **Exception (82% H)**

Engineering PCF 4.3 — BOMs + Routings	Materials PCF 4.1 + 4.4	Manufacturing PCF 4.3	Supplier External Participant
4.3.1 Develop Bill of Materials DOCUMENT M3: 40%	4.1.1 Develop Supply Chain Strategy KNOWLEDGE M3: 65%	4.3.2 Develop Production Schedule KNOWLEDGE M3: 65%	Ext Receive Purchase Order TRANSACTION M3: 35%
4.3.1 Define Production Routings KNOWLEDGE M3: 65%	4.1.2 Develop Demand Forecast KNOWLEDGE M3: 65%	4.3.3 Manage Work Orders TRANSACTION M3: 35%	Ext Confirm Materials Availability TRANSACTION M3: 35%
4.3.1 Manage Engineering Change Orders DECISION M3: 80%	4.1.3 Develop Inventory Plan DECISION M3: 80%	4.3.4 Execute Production TRANSACTION M3: 35%	Ext Deliver Materials / Components TRANSACTION M3: 35%
4.3.1 → Manufacturing (Release)	4.1.4 Develop Production Plan DECISION M3: 80%	4.3.5 Perform Quality Inspection DECISION M3: 80%	Ext → Materials (Goods Receipt)
	4.1.5 Manage Supply Chain Risk KNOWLEDGE M3: 65%	4.3.6 Manage Non-Conformance EXCEPTION M3: 82%	Ext Supplier Exception EXCEPTION M3: 82%
	4.4.1 Receive and Store Materials TRANSACTION M3: 35%	4.3.7 Record Production Output TRANSACTION M3: 35%	
	4.4.2 Manage Inventory TRANSACTION M3: 35%	4.3.7 → Materials (Inventory Update)	
	4.1.4 → Purchasing / P2P (4.2)	4.3.4 Production Exception EXCEPTION M3: 82%	
	4.1.2 Planning Exception EXCEPTION M3: 82%		

Value Chain Insight — Engineering anchors the process — BOMs, routings, and ECOs define what gets made and how. Materials is the operational hub: supply chain planning, inventory, logistics, and the handoff to Purchasing (P2P at PCF 4.2). Manufacturing executes, inspects, and closes the loop back to Materials inventory. Supplier delivers into Materials (Goods Receipt at PCF 4.4). P2Pr has the largest AI convergence gap at -25% — driven by transaction volume across production execution and materials management.